be safely ignored, but it can have implications when used as part of a more complex algorithm (see [29] and the discussion therein). The use of different formulations can result in differences in the accumulation of rounding error [167].[5]

2.2.3 Convergence and the Order of Accuracy

A typical integrator computes successive steps from the formulas

$$z_{n+1} = \mathscr{G}_h(z_n), \qquad z_0 = \boldsymbol{\zeta}.$$

Assume that $\mathscr{G}_h$ is a smooth map for all $h > 0$. The exact solution satisfies

$$z(t_{n+1}) = \mathscr{F}_h(z(t_n)).$$

To each $h > 0$ we may associate a finite set of phase space points $z_0, z_1, z_2, \ldots, z_\nu$; these represent the numerical solution at $t_0 = 0, t_1 = h, t_2 = 2h, \ldots, t_\nu = \nu h = \tau$.

Taking the difference of the numerical and exact solutions, we have

$$z_{n+1} - z(t_{n+1}) = \mathscr{G}_h(z_n) - \mathscr{F}_h(z(t_n)). \tag{2.9}$$

The first assumption is that $\mathscr{G}_h$ is an $\mathcal{O}(h^{p+1})$ approximation of $\mathscr{F}_h$ in the sense that there is a constant $K \geq 0$ and a constant $\Delta > 0$ such that, for $t \in [0, \tau]$, we have

$$\|\mathscr{F}_h(z(t)) - \mathscr{G}_h(z(t))\| \leq \bar{K} h^{p+1}, \qquad h < \Delta. \tag{2.10}$$

This assumption is usually verified by expanding the numerical and exact solutions in powers of h, using Taylor series expansions.

To tackle the question of the growth of local error, we still must make an important assumption on $\mathscr{G}_h$, namely that it satisfies a *Lipschitz condition* of the form

$$\|\mathscr{G}_h(\boldsymbol{u}) - \mathscr{G}_h(\boldsymbol{w})\| \leq (1 + hL)\|\boldsymbol{u} - \boldsymbol{w}\|, \qquad \boldsymbol{u}, \boldsymbol{w} \in D, h \leq \Delta. \tag{2.11}$$

The set D should be a domain containing the exact solution for $[0, \tau]$, and it is assumed that, for all $h \leq \Delta$ the numerical solution is also contained in D for $n =$

[5] Rounding error is the error introduced when numbers are forced into the finite word length representation in a typical digital computer. Adding together two "computer numbers," then rounding, results in another computer number. Rounding errors may accumulate in long computations, but in molecular dynamics they are normally dominated by the much larger "truncation errors" introduced in the process of discretization, that is, due to replacing the differential equation by a difference equation such as the Euler or Verlet method. For an example of the role of rounding error in the context of constrained molecular dynamics, see [237].

$0, 1, \ldots, \nu$. (This is a simplifying assumption the removal of which would require somewhat more intricate assumptions regarding the differential equations.)

With these assumptions in hand, begin from (2.9) and write

$$z_{n+1} - z(t_{n+1}) = \mathscr{G}_h(z_n) - \mathscr{G}_h(z(t_n)) + \mathscr{G}_h(z(t_n)) - \mathscr{F}_h(z(t_n)),$$

then take norms and use the triangle inequality and (2.10), (2.11) to get the following recurrent inequality for the error $\varepsilon_n = \|z_n - z(t_n)\|$:

$$\varepsilon_{n+1} \leq (1 + Lh)\varepsilon_n + \bar{K}h^{p+1}.$$

From this, the bound

$$\varepsilon_n \leq \frac{\bar{K}}{L} e^{Lnh} h^p, \qquad n = 0, 1, \ldots \nu, \tag{2.12}$$

follows by a straightforward calculation, for $h \leq \Delta$.

The assumption (2.10) that the local error is of order $p + 1$, $p > 0$, is termed the *consistency* of the numerical method. We say the method is *consistent of order p*.

The assumption (2.11) that the method does not increase the separation between two nearby trajectories by more than a factor of the form $1 + hL$ in each step is referred to as the *stability* of the method.

This result shows that a method which is consistent of order p and stable is convergent of order p.

Example 2.2 (The Verlet Method is 2nd Order) In this example, assume a single degree of freedom system, i.e. $q, p \in \mathbb{R}$, and take $M = 1$. The Verlet method can be written in the form of a map, as in (2.8), or, in slightly more detail, as

$$Q = q + hp + \frac{h^2}{2} F(q), \tag{2.13}$$

$$P = p + \frac{h}{2}\left[F(q) + F(q + hp + \frac{h^2}{2}F(q))\right]. \tag{2.14}$$

The first equation is already a polynomial, i.e. it is in the form of a series expansion in powers of h where the coefficients are functions of the starting point (q, p). The second equation may be written as a series expansion in powers of h as well:

$$P = p + \frac{h}{2}F(q) + \frac{h}{2}\left[F(q) + hF'(q)(p + \frac{h}{2}F(q)) + \frac{h^2}{2}F''(q)(p + \frac{h}{2}F(q))^2 + \ldots\right].$$

Note that the neglected terms will involve 4th (and higher) powers of h.